

Stateless Datacenter Load-balancing with **Beamer**

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Presented by Pascal Wiesmann

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Presentation overview

- What is the problem?
- How is it solved?
i.e. Contributions of the paper
- Evaluation

Goal: Better load balancing

- “[..] steer traffic to a given service across a dynamic set of backend machines.”
 - Fast
 - Robust (against attacks)
 - Stable (no big fluctuations)

Unhappy
Customers

== less money





26'969.-

Cisco 3850-32XS-E: 32
Port IPS Switch

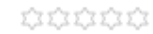
Cisco · Switch

Cisco 3850-32XS-E: 32 Port IPS Switch

Unbekannt

 In den Warenkorb

Cheap
hardware
== more money



24.-

Cisco SF110D-05

Cisco · Switch

Art-Nr 5831967

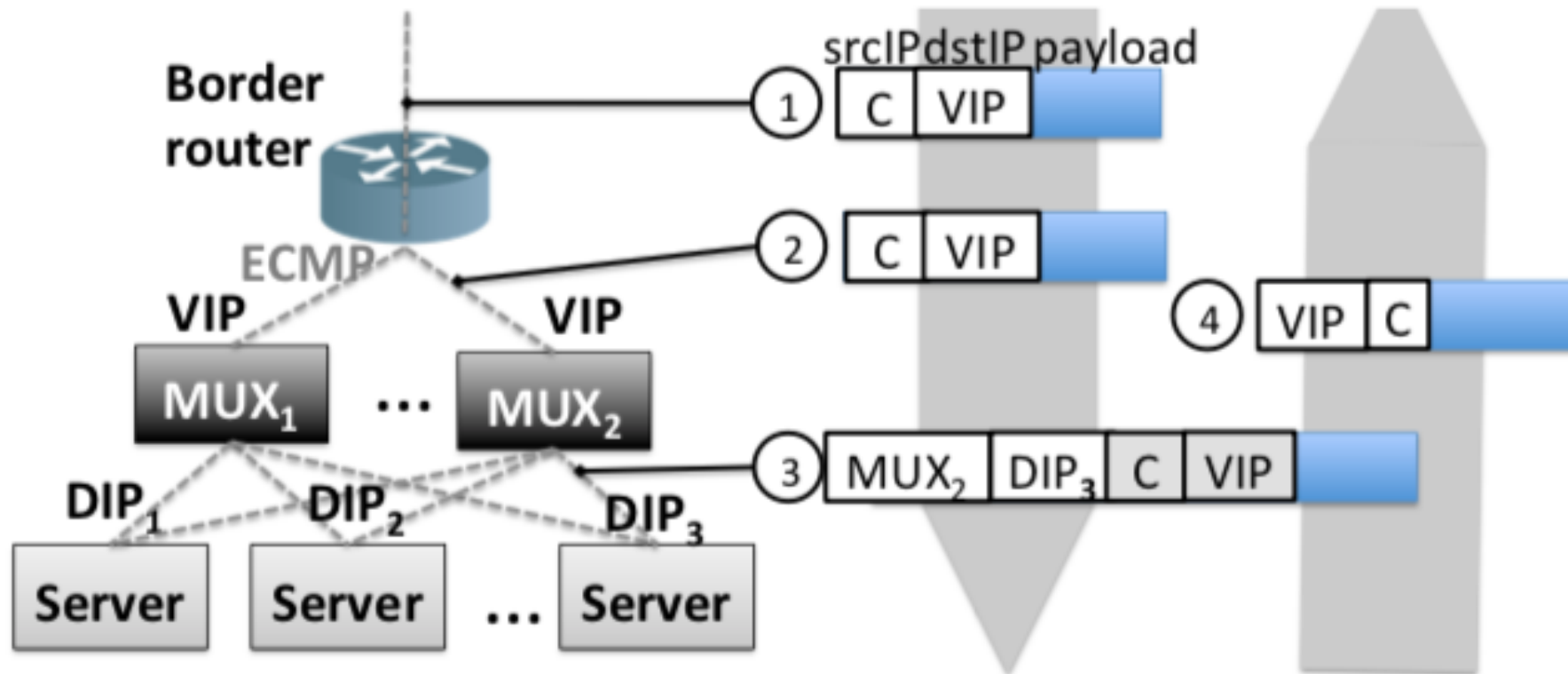
✓ Nur noch 3 Stück in unserem Lager



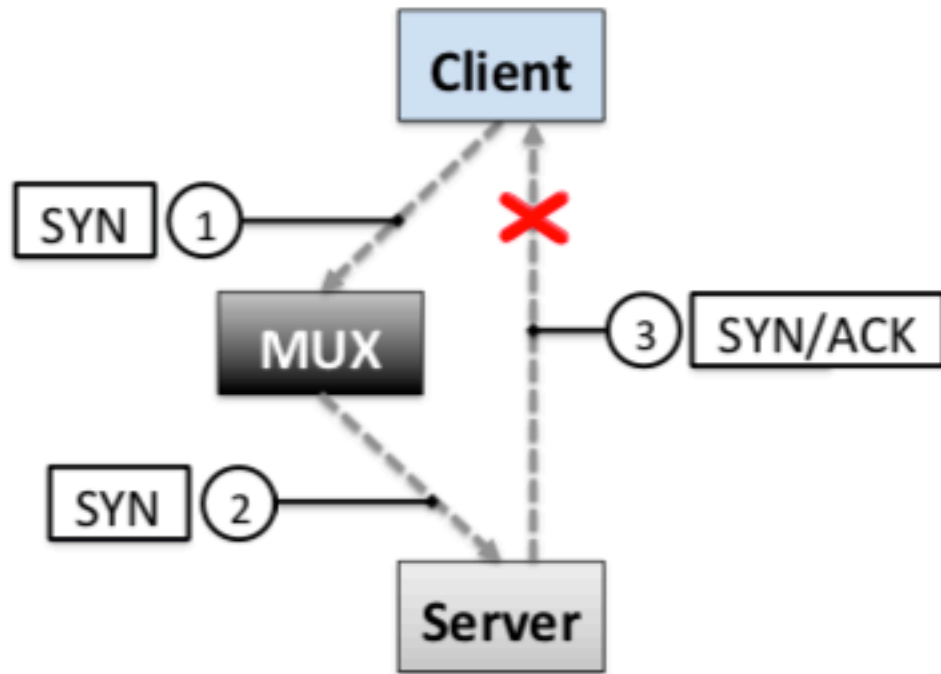
 In den Warenkorb

Load Balancing: Existing Approaches

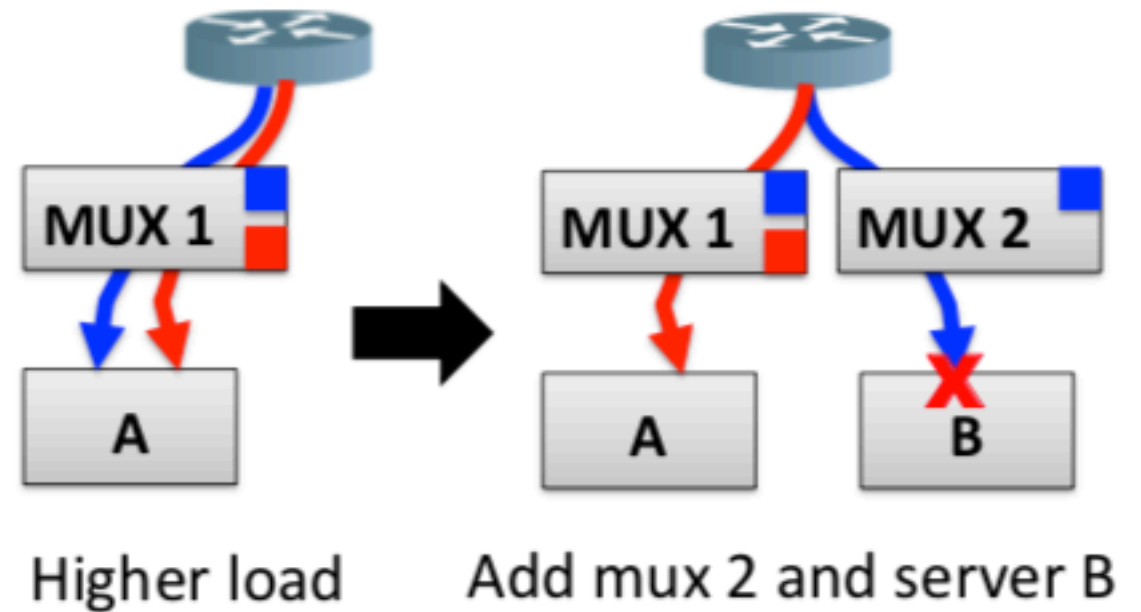
keeping per-flow state on the muxes



Problems of existing approaches



SYN floods



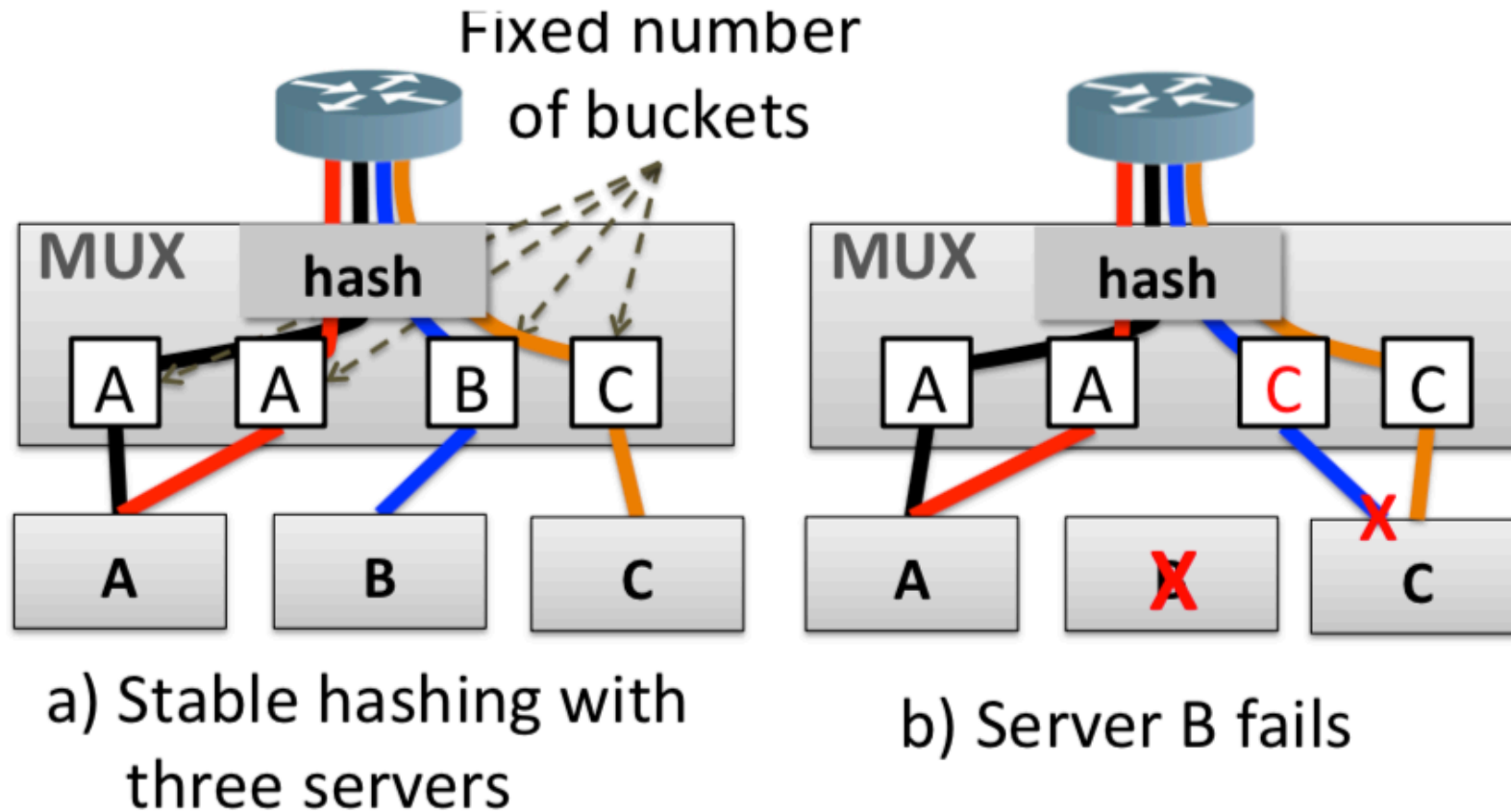
Scale-out

Beamer: stateless load-balancing

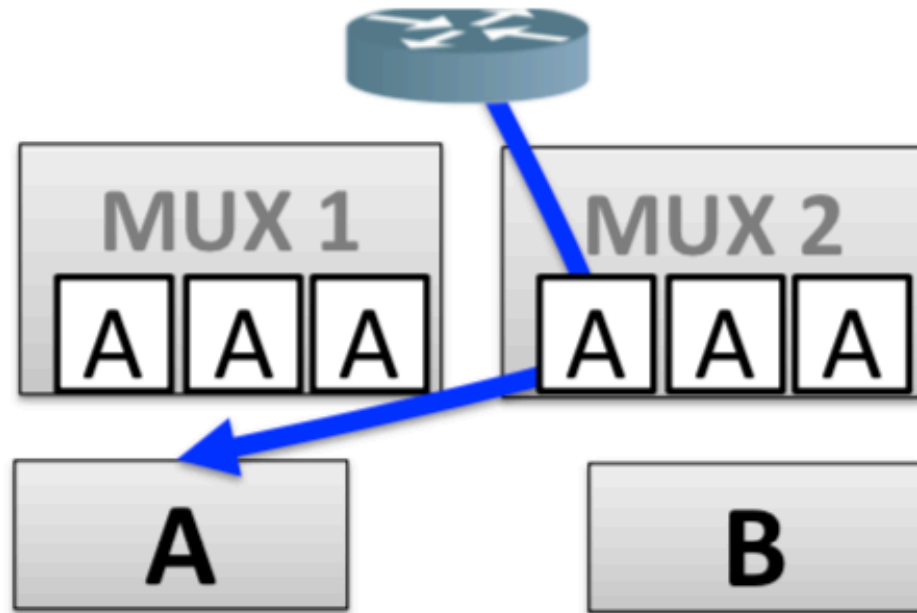
- Contributions:
 - Muxes without per-flow state
 - Stable hashing
 - Daisy chaining
 - Handling Multipath TCP

Stable Hashing

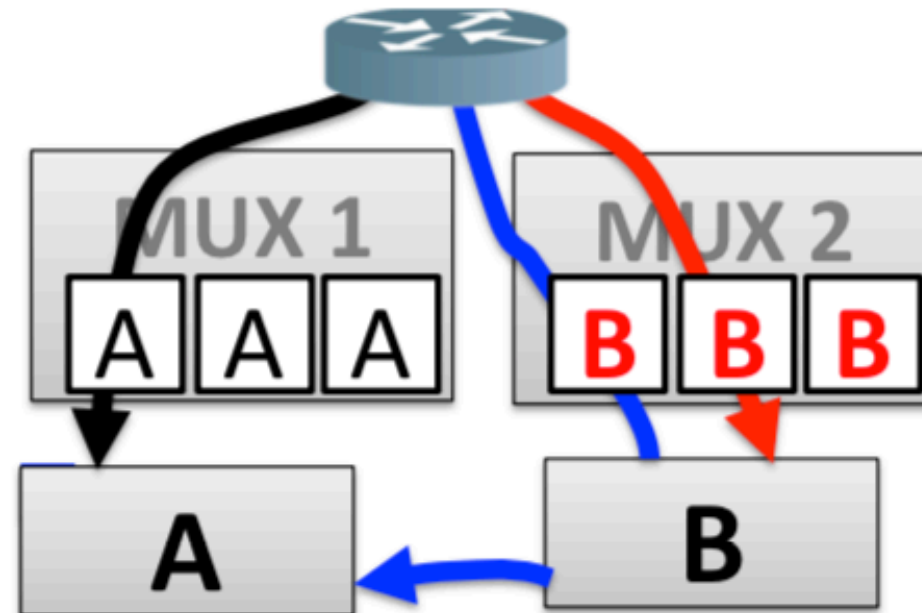
Trivial: $\text{hash}(5\text{tuple}) \% \#\text{servers}$
New: $\text{hash}(5\text{tuple}) \% \#\text{buckets}$
requires synchronization of bucket-to-server mapping between muxes!



Daisy Chaining

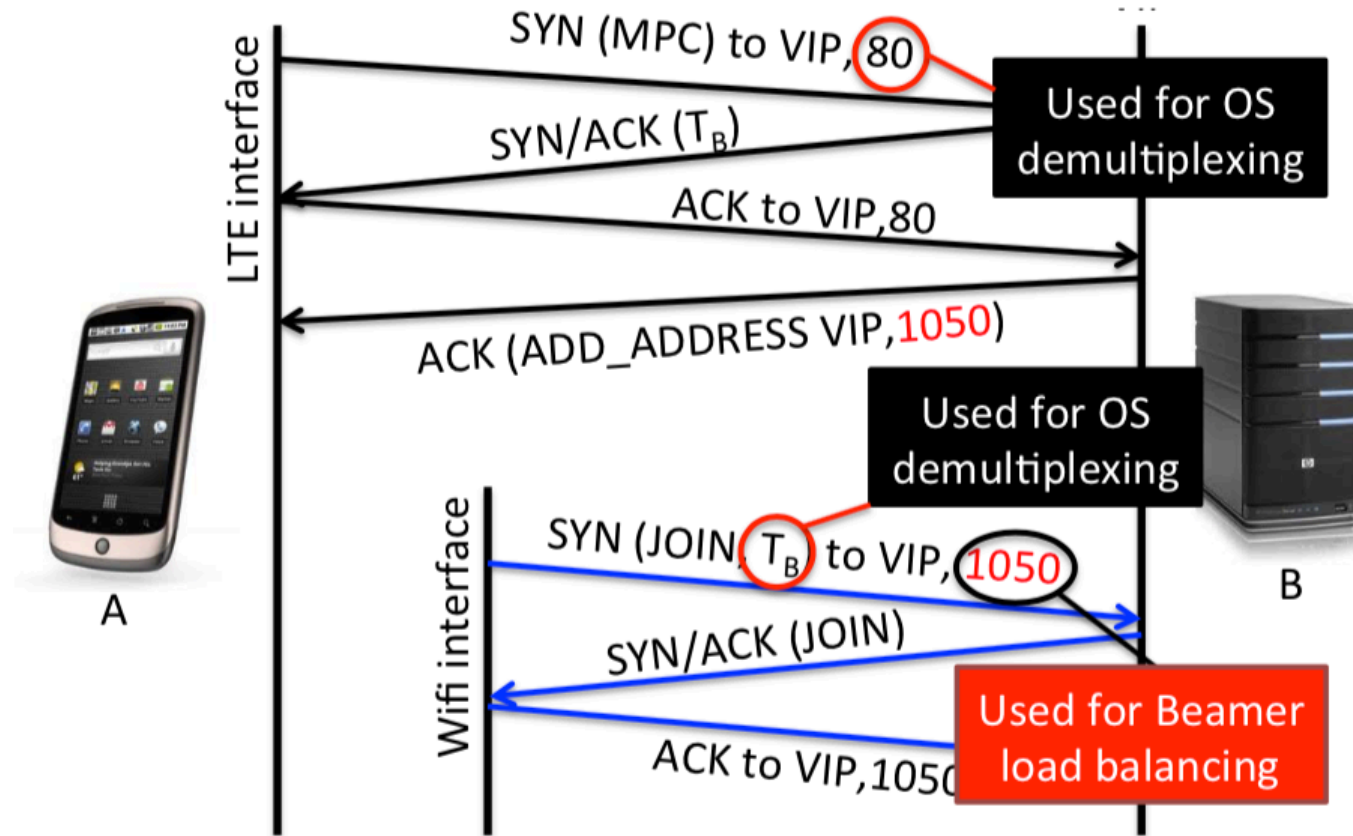


a) Buckets belong to server A



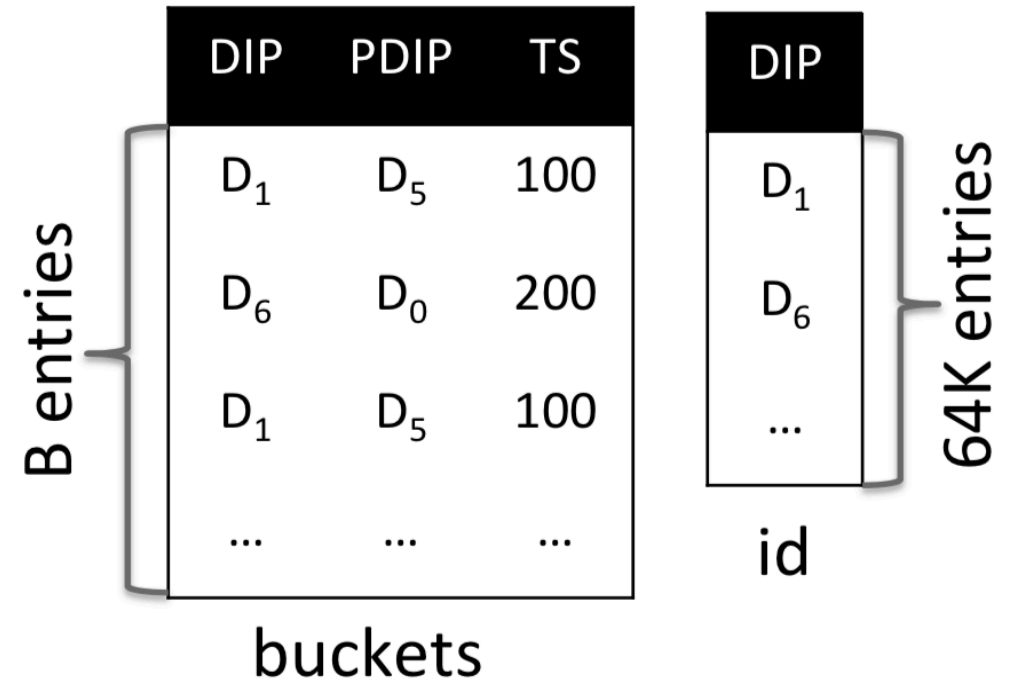
b) Buckets moved to B, Inconsistent mappings.

Multipath TCP (MCTCP)



Mux behavior and state

```
1 packet* mux(packet* p){
2   if (p->dst_port<1024){
3     gen = buckets.version
4     b = hash(5-tuple)%B;
5     dip = buckets[b][0];
6     pdip = buckets[b][1];
7     ts = buckets[b][2];
8
9     return encapsulate(mux,dip,pdip,ts,gen,p);
10  }
11  else {
12    dip = id[p->dst_port];
13    return encapsulate(mux,dip,p);
14  }
15 }
```



Benchmarks

“[...] our software implementation of Beamer is twice faster than Google’s Maglev, the state of the art software load balancer, and can process 40Gbps of HTTP uplink traffic on 7 cores.”

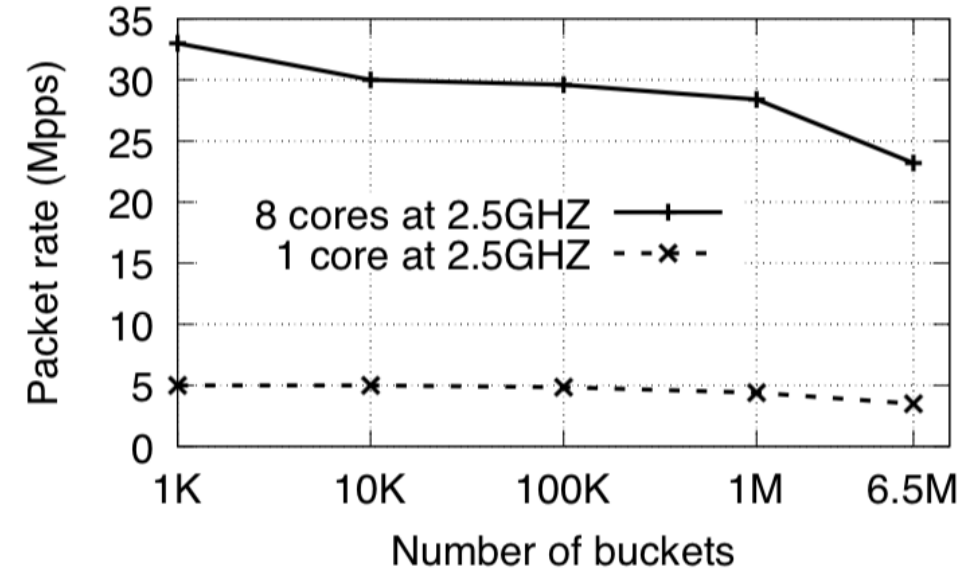
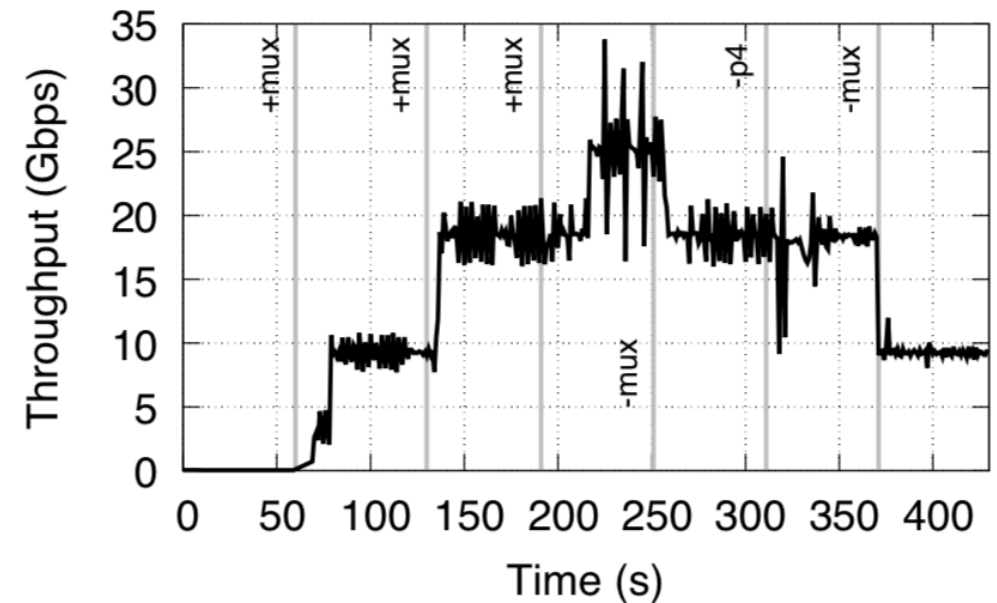
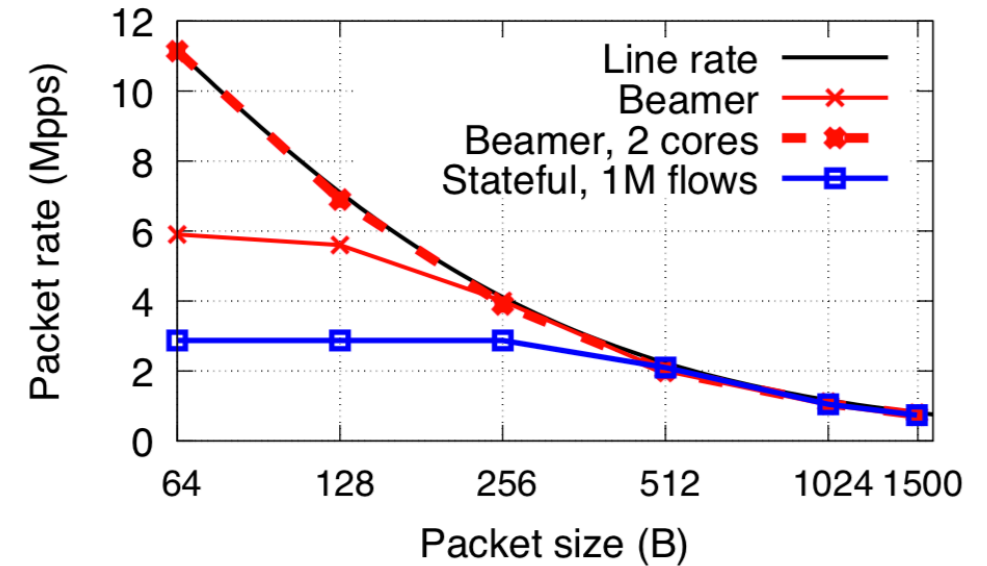


Figure 14: Forwarding performance (ten-core Xeon, 4 x 10Gbps). Beamer forwards 40Gbps with 128B packets.

More Benchmarks

- Faster than (own) stateful Baseline
- Mux latency: < 0.3ms
- Handles Mux churn smoothly



Benchmarks: MCTCP

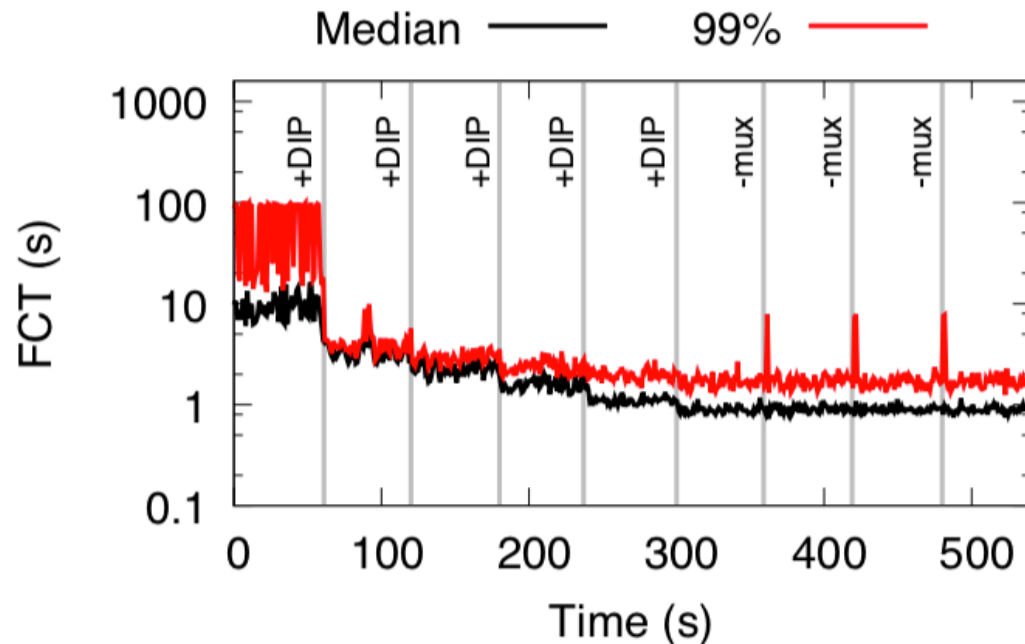


Figure 19: Flow completion times for MPTCP clients using Beamer

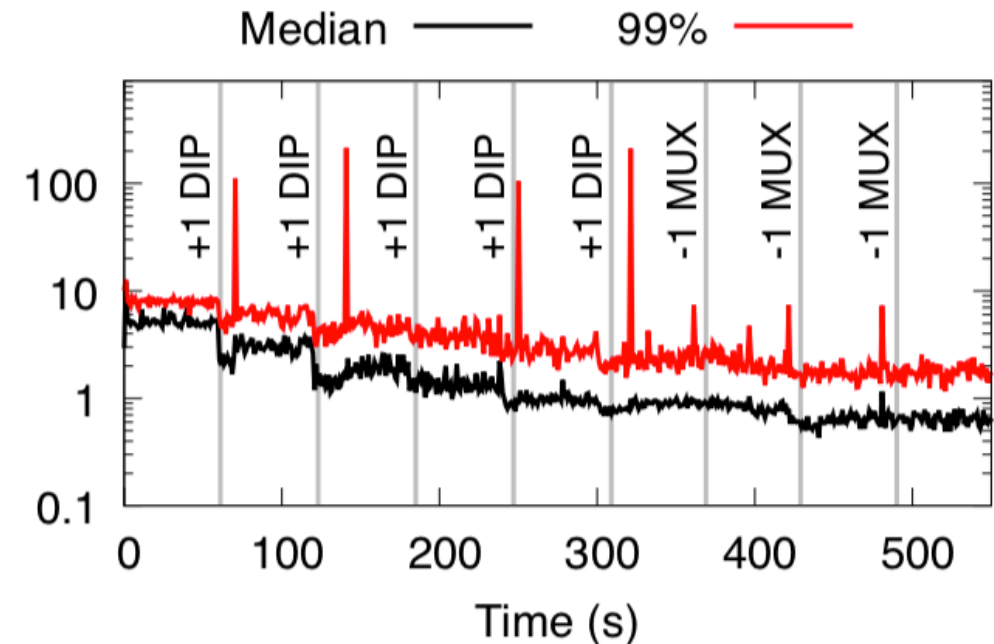


Figure 20: Flow completion times for MPTCP clients using Stateful.

Benchmarks: Controller update

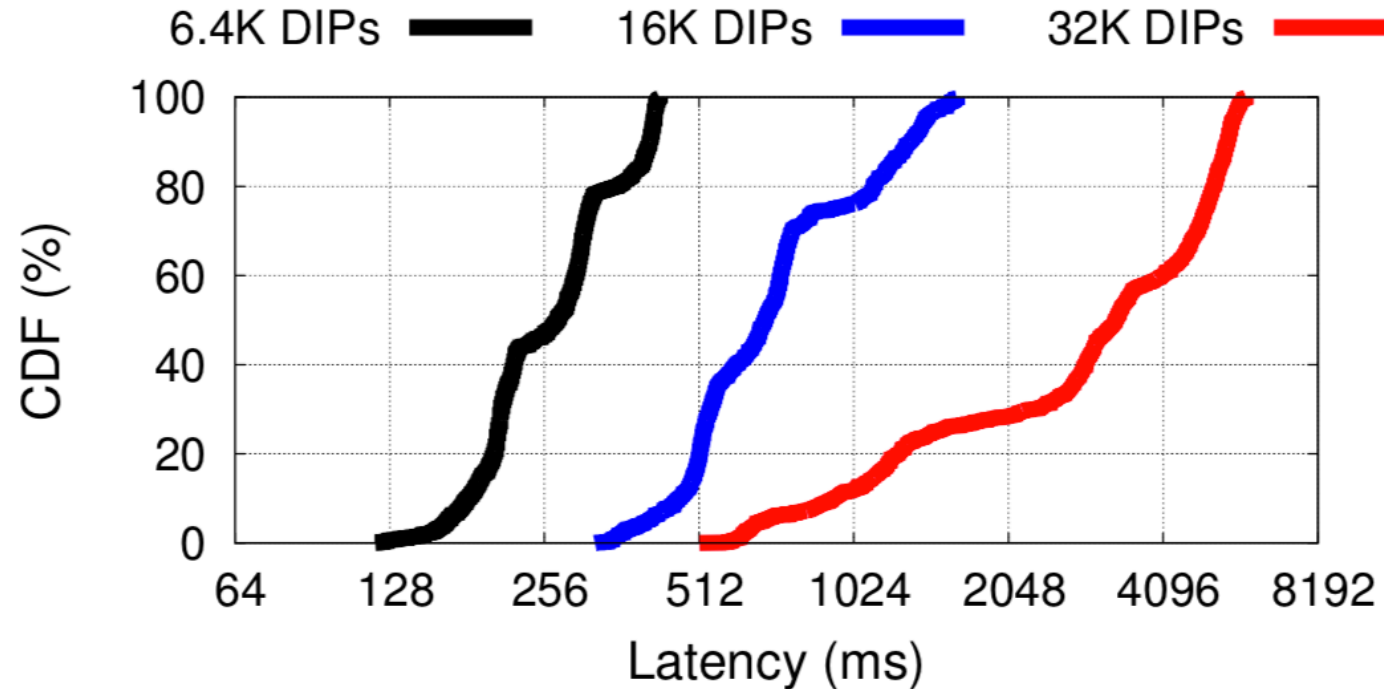


Figure 24: Time to propagate a controller update from ZooKeeper to all the muxes.

Beamer: Conclusion

- Pros:
 - Good connection affinity for TCP
 - Robust against SYN floods
 - Fast
 - Supports MPTCP
 - Open Source
- Possible Weaknesses
 - P4 not really tested
 - How important is the problem? How often does mux and server churn happen in practice?
 - Is the 40% speed improvement also observable in real world?

Paper Review

- Pro:
 - Well structured
 - Good explanations of the algorithms they introduce
- Con:
 - No “future work” section
 - Evaluation section unclear
 - Benchmark plots hard to understand
 - Only test against their own baseline implementation



That's all Folks!

Stateless Datacenter
Load-balancing with **Beamer**

Questions?